

30 Pulse and step experiments

Why Are We Learning This?

In the previous lecture, we introduced the concept of a **residence time distribution (RTD)** and discussed why real reactors do not behave exactly like ideal reactor models. The next question is: How do we actually measure an RTD?

The answer is tracer experiments. By introducing a harmless tracer into a reactor and monitoring when it leaves, we can determine how long molecules spend inside the reactor and quantify the reactor's mixing behavior. RTD experiments are among the most important diagnostic tools used to characterize real reactors.

By the end of this lecture, you should be able to:

- Explain the purpose of a tracer experiment.
- Describe the characteristics of a good tracer.
- Distinguish between pulse and step experiments.
- Define the RTD function $E(t)$.
- Define the cumulative distribution function $F(t)$.
- Explain the physical meaning of mean residence time.
- Explain the physical meaning of variance.

Course Roadmap

Introduction to CRE



General Balance Equation



Common Reactor Models
(CSTR, Batch, PFR)



Reaction Progress
(Conversion)



Graphical Reactor Design
(Inverse Rate/Levenspiel Plots)
↓
Rate Laws
(Kinetics)
↓
Equilibrium
(Reversible Reactions)
↓
Equilibrium Conversion
(Thermodynamic Limitations)
↓
One Reaction: Solving in terms of Reaction Progress
↓
One Reaction: Reactor Design
(CSTR, Batch, PFR, PBR)
↓
Catalytic Reactions
↓
Numerical Methods
(Python)
↓
Energy Balances
↓
One Reaction: Non-Isothermal Reactor Design
↓
Multiple Reactions
↓
Reactor Safety
↓
► Challenging the Well-Mixed Assumption ◀
(Residence Time Distributions)
↓
Challenging the Elementary Rate Law Assumption
(Reaction Mechanisms)

Big Idea

Inject Tracer



Measure Exit Behavior



Determine RTD



Understand Mixing

Tracer experiments allow us to measure how fluid actually moves through a reactor rather than how we assume it moves.

Residence time distributions (RTDs) are measured by dosing a "tracer" to the reactor.

A tracer

- Is non-
- Is easily
- Has physical properties
- Is often

What Makes a Good Tracer?

A tracer should behave like the reactor contents without affecting the reactor itself.

An ideal tracer: Does Not React, Does Not Alter Reactor Behavior, Is Easy to Detect, Moves With the Fluid

The tracer should faithfully follow the flow patterns inside the reactor so that its behavior reflects the behavior of the fluid being studied.

Physical Interpretation

Tracer



Acts Like a Molecule of Reactor Contents



Reveals Mixing Behavior

The tracer can be dosed in either a or experiment.

Pulse experiment: An amount of tracer is suddenly

The time needed to collect all of the tracer is used to determine the RTD.

Step experiment: A concentration of tracer is constantly

The time needed for the concentration of tracer exiting the reactor to equal that entering the reactor is used to determine the RTD.

Pulse vs Step Experiments

There are two common ways to introduce a tracer.

Pulse Experiment

A small quantity of tracer is injected almost instantaneously.

Quick Injection

↓

Watch Tracer Leave

↓

Measure RTD

The resulting data are used directly to obtain the RTD function $E(t)$.

Step Experiment

The inlet tracer concentration is suddenly changed and then held constant.

Constant Tracer Feed

↓

Measure Outlet Response

↓

Determine RTD

The resulting data are used to obtain the cumulative distribution function $F(t)$.

Key Difference

Pulse Experiment

↓

Measures $E(t)$

Step Experiment

↓

Measures $F(t)$

Important RTD statistics:

- ΔN : Amount of tracer leaving the reactor between t and Δt :
- $\frac{\Delta N}{N_0}$: The fraction of tracer/reactor contents that has a residence time between t and Δt :

- $E(t)$: Residence time distribution

The fraction of tracer/reactor contents leaving the reactor that has resided in the reactor for a time between t_1 and $t_2 = \int_{t_1}^{t_2} E(t) dt$

Also, $\int_0^{\infty} E(t) dt = 1$

What Does $E(t)$ Mean?

The residence-time distribution function $E(t)$ describes:

The Fraction of Material Leaving the Reactor That Has Spent Time t Inside the Reactor

In other words, $E(t)$ is a probability-density function for residence time.

Important Property

The total area under the RTD curve = 1

In other words:

$$\int_0^{\infty} E(t) dt = 1$$

because every molecule entering the reactor must eventually leave the reactor.

Physical Interpretation

Large $E(t)$



Many Molecules Leave At That Time

Small $E(t)$



Few Molecules Leave At That Time

- $F(t)$: Cumulative distribution function. Used for step experiments.

What Does $F(t)$ Mean?

$F(t)$ is the cumulative distribution function.

It represents: The Fraction of Material That Has Left the Reactor By Time t

Interpretation

If $F(t) = 0.60$ then 60% of the tracer has already exited the reactor by time t .

Relationship Between $E(t)$ and $F(t)$

$E(t)$



Probability Density

$F(t)$



Cumulative Probability

Both describe the same RTD but from different perspectives.

- t_m : Mean residence time
- σ^2 : Variance. Provides an indication of the "spread" in the data

What Do the Mean and Variance Tell Us?

Mean Residence Time

The mean residence time answers: How Long Does a Molecule Typically Remain In the Reactor?

It represents the average time spent inside the reactor.

Variance

Variance measures: How Spread Out The Residence Times Are

Low variance: Most Molecules Spend About the Same Time In the Reactor

High variance: Some Leave Early, Some Leave Late

Why Do We Need Statistics?

After collecting tracer data, we need a way to summarize the results. Questions include:

How long does a molecule typically remain in the reactor?

How broad is the RTD?

How non-ideal is the reactor?

RTD statistics provide quantitative answers to these questions.

Summary and Key Takeaways

Today we learned how residence time distributions are measured experimentally.

Key Ideas

- ✓ RTDs are measured using tracer experiments.
- ✓ A tracer should follow the fluid without affecting the system.
- ✓ Pulse experiments are used to obtain $E(t)$.
- ✓ Step experiments are used to obtain $F(t)$.
- ✓ $E(t)$ describes the distribution of residence times.
- ✓ $F(t)$ describes the cumulative fraction of material that has exited the reactor.
- ✓ Mean residence time measures the average residence time.
- ✓ Variance measures the spread in residence times and provides information about non-ideal behavior.

Most Important Idea

Tracer Experiment

↓

Measure Exit Response

↓

Determine RTD

↓

Characterize Mixing

↓

Understand Real Reactor Behavior

RTD experiments provide one of the most powerful methods for determining how fluid actually moves through a reactor and for quantifying deviations from ideal reactor behavior.